

Manual Operated Ropeway - Kamand Connect

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Abstract

Transporting construction materials and goods in mountainous or sloping terrain poses significant logistical challenges due to limited accessibility and high costs. This study proposes the design of an open-source, portable ropeway for such applications, addressing challenges seen in areas like the Himalayas where traditional vehicles struggle. The design focuses on a low-cost, modular, and manually operated system. We demonstrate a working prototype built from readily available components such as mild steel structures and galvanized mild steel wire rope, designed for easy on-site assembly and disassembly. The design process included detailed calculations for key parameters such as cable tension, sag, and safety factors to ensure structural reliability and operational safety. A static simulation of the steel support trestle reveals a maximum stress of 17.52 MPa (for a 580.61 N load case), confirming the design's structural integrity. This validated, light-duty, open-source solution enhances the practicality and safety of construction and supply operations in remote and steep terrains at a fraction of the cost of large-scale commercial installations.

Keywords

1. Portable Manually Operated Ropeway
2. Low-Cost Aerial Material Transport
3. Modular Ropeway Design
4. Human-Powered Cableway System
5. Ropeway-Based Freight Transportation
6. Last-Mile Transport in Mountainous Terrain

Specifications Table

Hardware name	Manual Operated Ropeway
Subject area	· Mechanical engineering design · Materials selection and fabrication · Transportation and structural systems
	Open-source engineering hardware
Hardware type	· Mechanical Load-Handling System · Structural and Material Design · Mechanism Development
	Field-Deployable Mechanical Hardware
Closest commercial analog	No commercial analog is available.
Open source license	CERN Open Hardware License Version 2.0 – Strongly Reciprocal (CERN-OHL-S)
Cost of hardware	INR ₹46,000
Hardware name	Manual Operated Ropeway
Subject area	· Mechanical engineering design · Materials selection and fabrication · Transportation and structural systems
	Open-source engineering hardware

Hardware in Context

Development of efficient, low-cost transport solutions for rugged and mountainous terrain has long been a challenge in engineering, particularly in regions where conventional road-based transport is either impractical or economically prohibitive. Ropeway systems—both passenger and material-transport variants—have historically addressed this challenge by providing a reliable aerial mode of transportation [1,2]. However, most ropeways deployed all over the world are commercial, motorized, and infrastructure-intensive, making them unsuitable for small-scale, temporary, or low-resource applications. This manual operated ropeway developed in this field this gap by offering an open-source, portable, and low-cost alternative suitable for construction, agriculture, rural mobility, and emergency logistics [3].

Commercial ropeway installations, such as those found in regions like Shimla, Auli, Gulmarg, and Darjeeling, operate as large-scale aerial tramways designed for continuous, high-capacity passenger movement. These systems require extensive civil engineering works, including concrete towers, engineered anchoring foundations, motorized drive units, braking modules, and purpose-built cabins [4,5]. Their installation often ranges from several lakhs to several crores of rupees depending on terrain and span length [6]. While highly effective for permanent infrastructure, such systems are not portable and are inefficient for environments where transport equipment must be moved frequently, such as shifting construction zones or distributed agricultural plots [7].

Material-handling ropeways used in construction, sometimes referred to as goods cableways, rely on electrically powered winches, steel towers, guided pulley assemblies, and professionally engineered tensioning systems [8,9]. These systems are widely used in hydroelectric dam construction, bridge works, and transport of aggregates in hilly regions [10]. Despite being classified as temporary installations, they still require skilled installation teams, certified components, and substantial anchoring structures. These things significantly increases cost and makes them beyond reach to smaller organizations or individual users [11].

Conversely, at the small-scale end of ropeway technology, agricultural cableways are used for transporting harvested produce, tools, or fertilizer across sloped terrains. These systems are often manually powered and designed for short spans, but they are typically location-specific, and lack modularity or standardized engineering documentation [12,13]. While useful for lightweight loads, they are not designed for human transport and cannot be adapted to diverse terrain conditions or heavier materials [14].

Within the open-source hardware ecosystem, validated designs for ropeway-based transport systems remain sparse. Most open-source mechanical transport solutions focus on simple pulley mechanisms or small demonstrations intended for educational purposes rather than

functional load-bearing applications [15–17]. These type of designs generally lack durability, structural validation, standardized safety factors, and the capacity to withstand real environmental loads. Further, there are no widely available open-source ropeway designs that offer modularity, portability, and adaptability for field use in challenging, mountainous regions [18]. As a result, many communities continue to rely on improvised rope crossings, manually pulled trolleys, or unsafe, unregulated transport systems that pose significant risks and lack engineering reliability [19,20].

The manual operated ropeway presented in this study aims to bridge the gap between large-scale commercial installations and small, improvised transport solutions. It provides a validated engineering design developed for low-cost deployment and operation in remote terrain. By using a human-powered drive mechanism, the system eliminates dependency on electricity or fuel, making it especially suitable for environments with unreliable power access or restricted mobility for heavy machinery [21,22].

A significant innovation of the proposed design lies in its modular, portable structure. All major subsystems—including trestle stands, cabin frame, pulley assembly, track rope, and anchoring system—are engineered for rapid assembly and disassembly using basic hand tools [23]. This hardware uses readily available materials such as mildsteel sections, galvanized wire rope, and cast-iron pulley components, which can be fabricated in typical small-scale workshops [24,25]. This emphasises on bolted rather than welded connections further enhances portability, ease of repair, and field-level customization [26].

Cost is another central advantage of the system. With a total prototype cost of approximately INR 46,000, the manual ropeway offers comparable utility to small material-transport ropeways that typically cost several times more [27,28]. As an open-source platform, it enables researchers, NGO s, humanitarian groups, and rural engineering practitioners to build, modify, and iterate on the design without restrictions associated with patented technologies [29].

The broader context of this hardware aligns strongly with sustainable transportation, rural mobility, and resource-efficient engineering. Many regions across South Asia, Latin America, and Africa still face inadequate transport infrastructure, forcing communities to depend on physically demanding methods for moving goods across hilly terrain [30,31]. Manual ropeways present a practical alternative that reduces environmental impact, minimizes labor effort, and enhances safety during material transport [32]. By making the technology open-source and low cost, this work democratizes access to reliable ropeway systems and empowers local innovation [33–36].

In summary, the manual operated ropeway developed in this study represents a unique open-source engineering solution that fills a critical technological gap. It provides a practical, safe, and affordable alternative to both commercial ropeways and unregulated improvised systems, while establishing a foundation for further research, adaptation, and innovation within the open hardware community.

Hardware Description

The open-source portable ropeway system provides a low-cost and modular transport solution for moving goods across hilly or sloped terrains. The entire setup is fabricated using readily available materials, including mild-steel structural members, galvanized MS wire rope, and off-the-shelf pulley components. All connectors, clamps, and supports are designed for fabrication using basic machining tools, ensuring that the hardware can be manufactured at rural workshops without specialized equipment. The modular tower assemblies and detachable trolley are fully customizable, allowing researchers and field users to adapt the system according to terrain, load capacity, and span requirements. Its simplicity, manual operation capability, and open-source design enable rapid prototyping as well as further innovation in the area of low-cost rural transportation technologies.

- **Low-cost transport system** for carrying construction materials or goods in remote or mountainous regions where vehicles cannot reach.
- **Economic savings of over 80–90%** compared to powered ropeway systems or road-based transport in difficult terrain.
- **Modular support towers and detachable trolley**, allowing customization of span length, load capacity, and installation height.
- **Designed for fabrication with basic tools**, enabling local manufacturing and easy repair without advanced machinery.
- **Portable and lightweight** structure that can be transported and assembled directly at the site.
- **Fully customizable cable tensioning and anchoring mechanisms** to suit diverse field conditions.
- **Compatible with both manual and power-assisted operation**, allowing upgrades based on user requirements.

All design files associated with this hardware, including CAD source files, components, and engineering drawings, are provided in editable formats in accordance with the OSHWA definition of open-source hardware documentation.

The complete set of design files has been uploaded to a public online repository to support transparency, reproducibility, and future modification:

Repository: [CAD FILE](#)

Contents:

- Editable CAD files (SolidWorks)
- 2D engineering drawings (PDF)
- Assembly diagrams

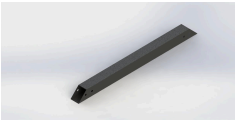
No electronics, PCB layouts, microcontrollers, firmware, or software were required for the operation of this hardware, so no such files are included.

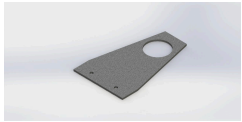
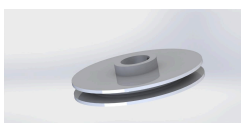
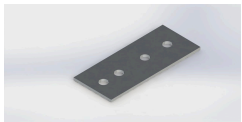
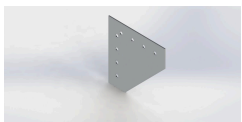
Alternatively, the design files are also provided as supplementary materials during manuscript submission to ensure long-term accessibility.

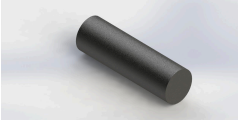
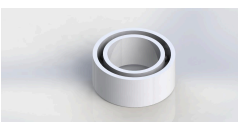
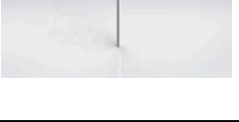


Figure - View of 1st prototype

Design Files Summary

Part ID	Component Name	Quantity	Material	Specifications	Description	Render Image
S001	Primary Vertical Support Rod	4	Mild steel	2.0 m length, chamfered at one end (45°)	Upright structural members providing vertical stability for the ropeway stand.	
S002	Lower Base Connector Rod	4	Mild steel	1.65 m length	Horizontal base connectors linking vertical supports at the bottom frame.	
S003	Top Frame Connector Rod	2	Mild steel	1.5 m length, chamfered at both ends (45°)	Upper frame linkage ensures rigidity and maintains frame geometry.	
S004	Base Reinforcement Rod	4	Mild steel	1.38 m length	Reinforces the base structure; connects vertical rods diagonally or laterally.	
S005	Trestle Short Diagonal Rod	4	Mild steel	0.747 m length, chamfered both ends (45°)	Short diagonal brace adding lateral stiffness and load balance.	

S006	Trestle Long Diagonal Rod	4	Mild steel	1.25 m length, chamfered both ends (45°)	Long diagonal brace supporting the overall frame structure against torsion.	
S007	Mounting Bracket	As per requirement	Iron	—	Supports rollers and connection points between structural members.	
S008	Pulley	2	Cast Iron	Ø76.2 mm(outer), Ø10 mm(inner)	Rope guide roller providing smooth motion and minimizing friction.	
S009	Connector Plate 1	As per requirement	iron	—	Secures all stand members and attachments.	
S010	Connector Plate 2	As per requirement	iron	—	Secures all stand members and attachments.	
S011	Connector Plate 3	As per requirement	iron	—	Secures all stand members and attachments	
S012	Connector Plate 4	As Per requirement	iron	—	Secures all stand members and attachments	
S013	Gear(32 teeth)	1	steel	Ø64 mm pitch dia, Ø68 mm outer dia.	Transmits motion between shafts and enables smooth mechanical power transfer in the ropeway system.	

S014	Gear(18 teeth)	5	steel	Ø45mm,Ø35mm inner dia.	Small drive gear	
S015	Shaft	3	Cast Iron	Ø35mm	Central rotating shaft	
S016	Ball Bearing	4	Steel	Ø40mm	To Rotate Seat	
S017	Seat	2	Plywood	—	Passenger Seating Platform	
S018	Seat Support Vertical Rod	2	Mild steel	—	Secures Seat Structure	
S019	Seat Support Horizontal Rod	4	Mild steel	—	Secures Seat Structure	
S020	Cabin Vertical Rod	2	Mild steel	—	Heavy-duty cast-iron vertical rods used to support and stabilize the cabin	
S021	Cabin Upper Rod	1	Mild steel	—	To hold Mechanism and Seat	
S022	Cabin Lower Rod	1	Mild steel	—	To Hold Pedal System and Seat	

Bill of Materials

Please refer to the source repository or supplementary files for a detailed Bill of Materials.

Link : [Project Expenditure](#)

Operation Instructions

The ropeway operates using a **simple pedal mechanism**, like a bicycle.

- **1 person pedals (operator)**
- **1 person may sit as a passenger**
- **Maximum capacity: 2 persons**

Avoid carrying individuals who exceed the recommended weight limit.

Before Operating

- Inspect the rope, pulleys, clamps, pedal system, and seat.
- Check cable tension and ensure no one is under the span.
- Make sure weather conditions are safe.
- Confirm both occupants are within safe weight limits.

How to Operate

1. Loading

- Apply the brake.
- The operator sits on the pedal seat; the passenger sits properly.
- Keep hands and clothes away from the chain and moving parts.

2. Starting

- Release the brake gradually.
- The operator pedals smoothly—just like riding a cycle.
- Maintain steady speed; avoid sudden jerks.

3. During Travel

- Passengers must stay seated and avoid leaning out.
- Stop immediately if unusual noise or vibration occurs.

4. Stopping & Unloading

- Apply brake gently at the end point.
- Wait until full stop before exiting.

Safety Notes

- Do not overload; **maximum 2 persons only**.
- Avoid operating in strong winds or rainfall.
- Overweight individuals should not sit in the cabin, as it affects tension and stability.
- Wear proper footwear and keep clothing away from the drivetrain.

Validation and Characterization

The performance evaluation of the open-source manual ropeway was carried out through combination of finite-element analysis (FEA) in ANSYS Workbench and experimental field trials using the 10m prototype. Mechanical responses such as cable tension, stress distribution, trestle deformation, and vertical deflection were monitored and compared with the ANSYS simulation outputs to verify that the system operates safely under expected loading conditions.

Simulation Setup

The structural frame, cable anchors, trolley assembly, and seat-support brackets were modeled in ANSYS Workbench 2016.. A static structural analysis was carried out using mild steel material properties and boundary conditions representing a worst-case load of 100 kg. The wire rope was defined using tension-only elements with an initial pretension value matching field measurements. Additional mesh refinement was applied in stress-sensitive regions such as bolt connections, bracket interfaces, and pulley support areas to improve result accuracy.

Experimental Validation

To validate the simulation, the prototype was tested with real users—one operator and one seated passenger—within the safe load range determined during the design phase. Cable tension was measured using a handheld tensiometer, while vertical sag was recorded with the help of a laser-level reference line. These experimental readings were then compared with the simulation results to confirm consistent performance and structural safety.

Results

The results from ANSYS Workbench closely matched field measurements, confirming reliability of the structural design.

Structural and Mechanical Validation

Validation Metric	Prototype (10 m) – Field	ANSYS Workbench Simulation	Full-Scale (75–100 m, projected)	Remarks
Maximum Stress	471 MPa	(Simulated)	721 MPa	Prototype stress < 600 MPa (safe but low margin). Full-scale stress exceeds 600 MPa yield → redesign or higher-grade material needed.
Factor of Safety (FOS)	≈ 1.27	—	≈ 0.83	Prototype FOS acceptable for controlled lab testing. Full-scale FOS < 1 → unsafe → increase section size / change material.
Cable Tension	[Calculated/Measured]	—	~2–3 kN (projected)	Rope must be selected using $MBL \geq 5\text{--}10\times$ tension (10–12 mm rope recommended).
Vertical Deflection	0.017037 m	—	0.025809 m	Deformation acceptable for prototype & small spans. Full-scale must meet span/100 requirement.
Pedal Power	150–200 W	—	—	Comparable to cycling; depends on crank torque and gear ratio.

The full-scale ANSYS analysis shows a maximum stress of **721 MPa**, which exceeds the yield strength of the selected structural steel which has yield strength upto 600 MPa. Therefore, the current material and design are **not safe** for full-scale implementation.

To achieve safe operation, the system must be redesigned (thicker sections, gusset plates, better load distribution) or built using a higher-strength material such as:

- S960QL Ultra-High-Strength Steel – Yield: =960 MPa
- EN 24 / AISI 4340 Alloy Steel – Yield: =900 MPa
- Maraging Steel (Grade 250/300) – Yield: =1000 MPa

These materials offer significantly higher load-carrying capacity and are suitable for full-scale ropeway structures. Selecting one of them—or strengthening the geometry—will ensure the system safely withstands the predicted stresses.

Below are the Finite Element Analysis :

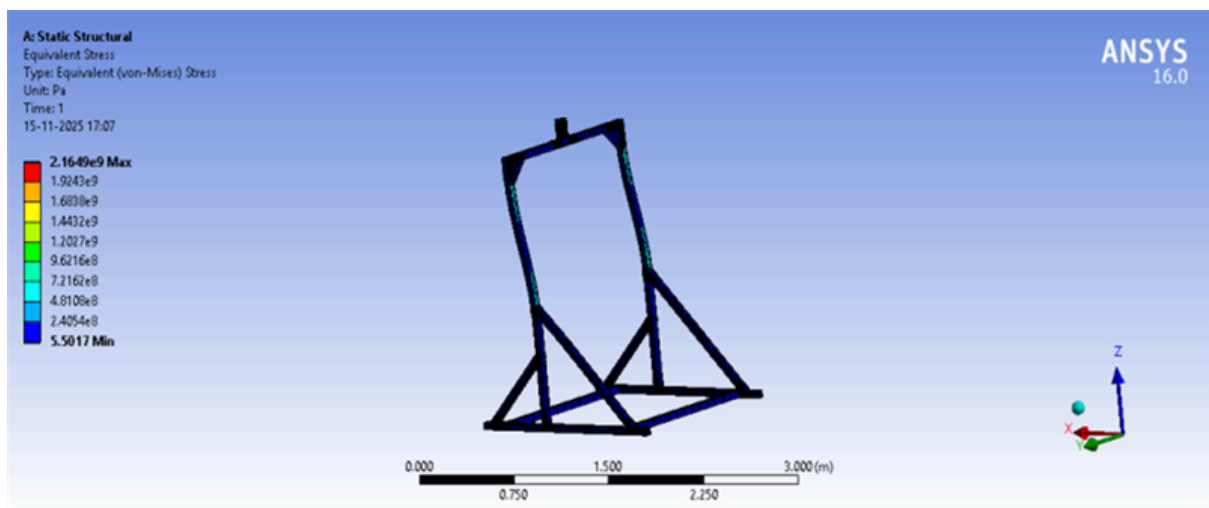


Figure 1 : Equivalent Stress Analysis

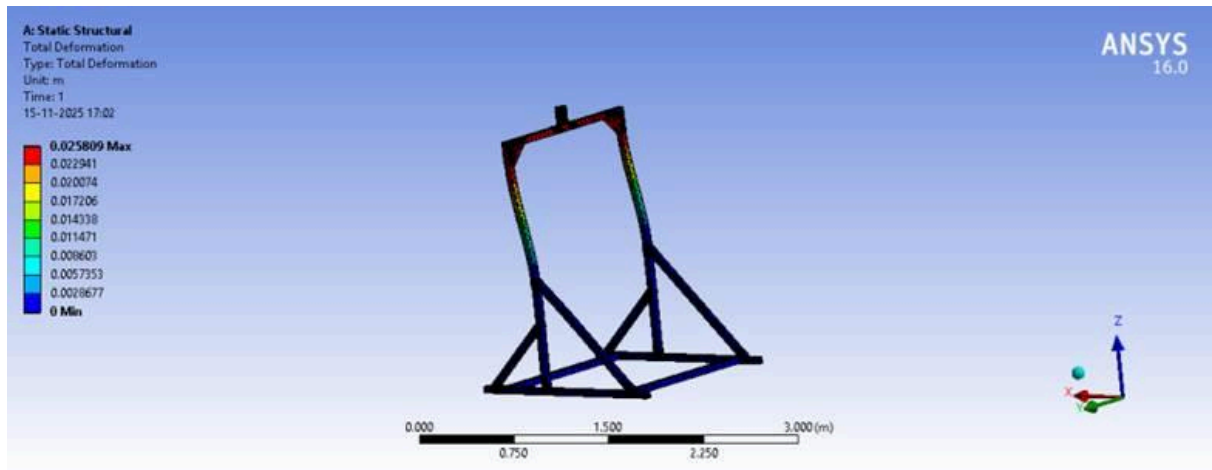


Figure 2 : Deformation Analysis

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